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ORIGINAL PAPER

Environmental drivers of local and temporal variations in the community of oak-associated borers (Coleoptera: Buprestidae)

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Abstract

Forest health is being challenged worldwide due to changing disturbance regimes. These changes could promote the abundance and diversity of secondary forest pests, like the Agrilinae (Coleoptera: Buprestidae) borers, which are frequently involved in oak decline in Europe. We assessed the contribution of environmental factors to local abundance and temporal dynamics of oak-dwelling borers in France. We set up green multi-funnel traps in oak stands exhibiting a gradient of decline across six forests in 2021 and monitored the traps for six consecutive years in one forest. The community of borers was homogeneous among the forests and included several Mediterranean species. Increasing severity of decline at the plot and stand scales enhanced the abundance and species richness of the borers and influenced the composition of their community. Tree density, longitude and spring temperature also influenced the abundance of several species and of the community. Our multi-year survey indicated that inter-annual fluctuations in decline level weakly correlated with those of borer species. Most of the species experienced a major crash in 2017, suggesting that extreme climate events such as a warm early spring followed by a late frost can decimate borer populations. This may explain why borer damage has recently been relatively limited in France.

Keywords *Agrilus* · *Coraebus* · *Quercus* · Agrilinae · Late frost · Drought

Introduction

Climate change exacerbates the effects of several types of disturbance in forest ecosystems, such as droughts, wildfires and windstorms (Seidl et al. 2017), leading to detrimental effects on forest health worldwide, including large-scale decline and dieback (Allen et al. 2010; McDowell et al. 2020; Senf and Seidl 2021). Large amounts of weakened trees may promote outbreaks of opportunistic pests, like bark beetles in conifer forests (Pineau 2017; Biedermann et al. 2019).

In broadleaved forests, borers in the Agrilinae (Coleoptera: Buprestidae) group are prominent aggravating agents of forest decline worldwide (Sallé et al. 2014; Haack and Petrice 2019). Their larval development in the cortical tissues of weakened trees and shrubs can cause stem girdling and disrupt sap flow, thereby affecting host vigour and even jeopardising survival (Vansteenkiste et al. 2004). In Europe, *Agrilus biguttatus* (Fabricius) and *Agrilus viridis* (Linnaeus) are frequently involved in oak and beech decline and dieback, respectively (Sallé et al. 2014; Brown et al. 2017; Brück-Dyckhoff et al. 2019; Macháčová et al. 2022). Several other species of Agrilinae have also been associated with oak decline, such as *Agrilus angustulus* (Illiger), *Agrilus sulcicollis* Lacordaire, *Coraebus undatus* (Fabricius) and *Coraebus fasciatus* (Villers) (Du Merle and Attié, 1992; Vansteenkiste et al. 2004; Sallé et al. 2014, 2020; Macháčová et al. 2022). Consequently, large-scale decline and dieback of oak forests should promote both the abundance and diversity of oak-associated borers. In addition, Agrilinae are thermophilous and heliophilous species, and their distribution is limited by their thermal requirements (Du Merle and Attié, 1992; Cardenas and Gallardo 2013; Reed et al. 2018). Thus, in temperate areas, global warming may directly promote their populations, especially those of Mediterranean species. For

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example, several species of borers have recently extended their geographical distribution range northwards in Europe (Pederson and Jørum 2009; Buse et al. 2013; Brown et al. 2015). Climate change may consequently have both direct (i.e. through survival and developmental effects) and indirect (i.e. by increasing the number of favourable, weakened hosts) positive effects on borers in Europe, which gives rise to fear their increasing economic and ecological incidence (Moraal and Hilszczanski 2000; Sallé et al. 2014).

According to the French Forest Health Department (DSF) and the National Forest Inventory (IFN), the overall forest sanitary condition in France has been deteriorating for the last few years, including in oak forests (Drain et al. 2021). Several oak forests exhibited increased tree dieback and mortality following the severe summer droughts in 2018, 2019 and 2020 (Saintonge and Goudet 2020). Nevertheless, despite the increasing number of weakened oaks, records of borer damage in the DSF database remained relatively stable over the same period (Sallé et al. 2022), while massive colonization of weakened oaks has been reported recently in Germany (Hurling et al. 2023). This suggests that, in addition to oak sanitary conditions, several environmental factors may affect the population dynamics of borer species at local scales, modulating their contribution to the decline process.

While there is growing concern about the ecological and economic impacts of Agrilinae on forest ecosystems, few studies have investigated the community's composition, its temporal variations and the factors underpinning these variations (Redilla and McCullough 2017; Sallé et al. 2020). Basic knowledge of the population dynamics of borers in their native range would be useful to optimise monitoring their populations either during forest decline and dieback events or in the context of biological invasions. Some European *Agrilus* species have already become established outside Europe, mostly following accidental introductions (Jendek and Grebennikov 2009), and several other Agrilinae have already proven to be particularly aggressive invasive species, acting as primary pests on native hosts in their introduction area (Coleman and Seybold 2011; Herms and McCullough 2014; Bozorov et al. 2019).

The main objective of the present study was to identify the environmental factors which influence the community of oak-associated borers in temperate forests. We assessed the effects of climatic, geographic and stand-related factors on the abundance and species richness of oak-dwelling borers. We hypothesised that declining oak stands may shelter more abundant and diverse communities of borers than do healthy ones, and that both abundance and diversity should increase over time with decline severity in affected forests. As Agrilinae are thermophilous species, we expected more diverse and more abundant communities under warmer conditions and/or at lower latitudes. We also expected that low tree density at the stand scale, leading warmer summer conditions (Díaz-Calafat et al. 2023), may also promote borer populations. In addition, we assessed whether the population level and diversity of oak borers could contribute to local oak decline severity. To verify these hypotheses, we sampled oak-associated borers in six distant forests and, in one forest, maintained monitoring for six consecutive years. Our second aim was to provide information regarding community composition and phenology to facilitate monitoring these European species.

Material and methods

Study area

The study was conducted in six lowland forests located in France (Table 1; Fig. 1). The two dominant tree species in these forests were *Quercus robur* L. and *Quercus petraea* (Matt) Liebl. All of the forests are managed by the National Forestry Office (ONF) for the production of high-quality oak timber, and consequently, all the sites undergo similar forest management strategies. In each forest, we visually selected several oak stands along a gradient of decline from healthy to severely declining, based on their crown condition (see the section on plot and stand characteristics below), in order to obtain an equivalent number of severely declining, moderately declining and healthy stands.

The Vierzon forest had been affected by several decline and dieback events since 1920 and was still experiencing oak mortality in several stands during the survey period (Sallé et al. 2020; Sardin et al. 2022). Here, the prominent decline-inducing factors were defoliation caused by powdery mildew and severe droughts (Douzon 2006; Marçais and Desprez-Loustau 2014), and the major aggravating agents were Armillaria honey fungi, *Armillaria mellea* (Vahl) Kumm. and *A. biguttatus* (Douzon 2006). In the Chantilly forest, the sandy soil promoted both cockchafer, *Melolontha hippocastani* (Fabricius), outbreaks and droughts. As a result, this forest has experienced severe decline for the last 20 years (Le Bouler 2023). The health condition of the Marcenat, Châteauroux and Orléans forests was evaluated during a national survey of oak forests conducted in 2020 (Saintonge and Goudet 2020). At that time, the Marcenat and Châteauroux forests were experiencing a decline, subsequent to the recurrent summer droughts of 2018 and 2019, while the Orléans forest was healthier (Saintonge and Goudet 2020). Likewise, the Boulogne forest was considered healthy.

Beetle sampling and identification

The data were collected during two successive projects: the first from 2016 to 2018 and the second from 2019 to 2021;

Table 1 Characteristics of the monitored forest

Forest name	Monitoring year(s)	No. of stands	No. of traps	Lat	Long	Dominant trees	Understory	Surf. (ha)	Sanitary condition
Boulogne	2021	6	12	47.57	1.53	Oaks (86%)	Hornbeam, pines	4,068	Healthy
Chantilly	2021	3	6	49.17	2.51	Oaks (59%), lime (19%),	Hornbeam, chestnut	6,314	Severe decline
Châteauroux	2021	6	12	46.74	1.72	Oaks (100%)	Hornbeam, beech, holly	5,322	Moderate decline
Marcenat	2019 and 2021	3	5	46.23	3.37	Oaks (71%), pines (11%)	Hornbeam, lime	2,066	Moderate decline
Orléans	2019 to 2021	9 (2019)	18	47.87	2.41	Oaks (47%), pines (51%)	Hornbeam	34,699	Slight decline
		6 (2020, 2021)	12						
Vierzon	2016 to 2021	11 (2016–2018)	11	47.46	2.19	Oaks (61%), pines (31%)	Hornbeam	7,532	Severe decline
		9 (2019, 2020)	18						
		6 (2021)	12						

Lat latitude, *Long* longitude, *No.* number, *surf.* total surface area of the sampled forest

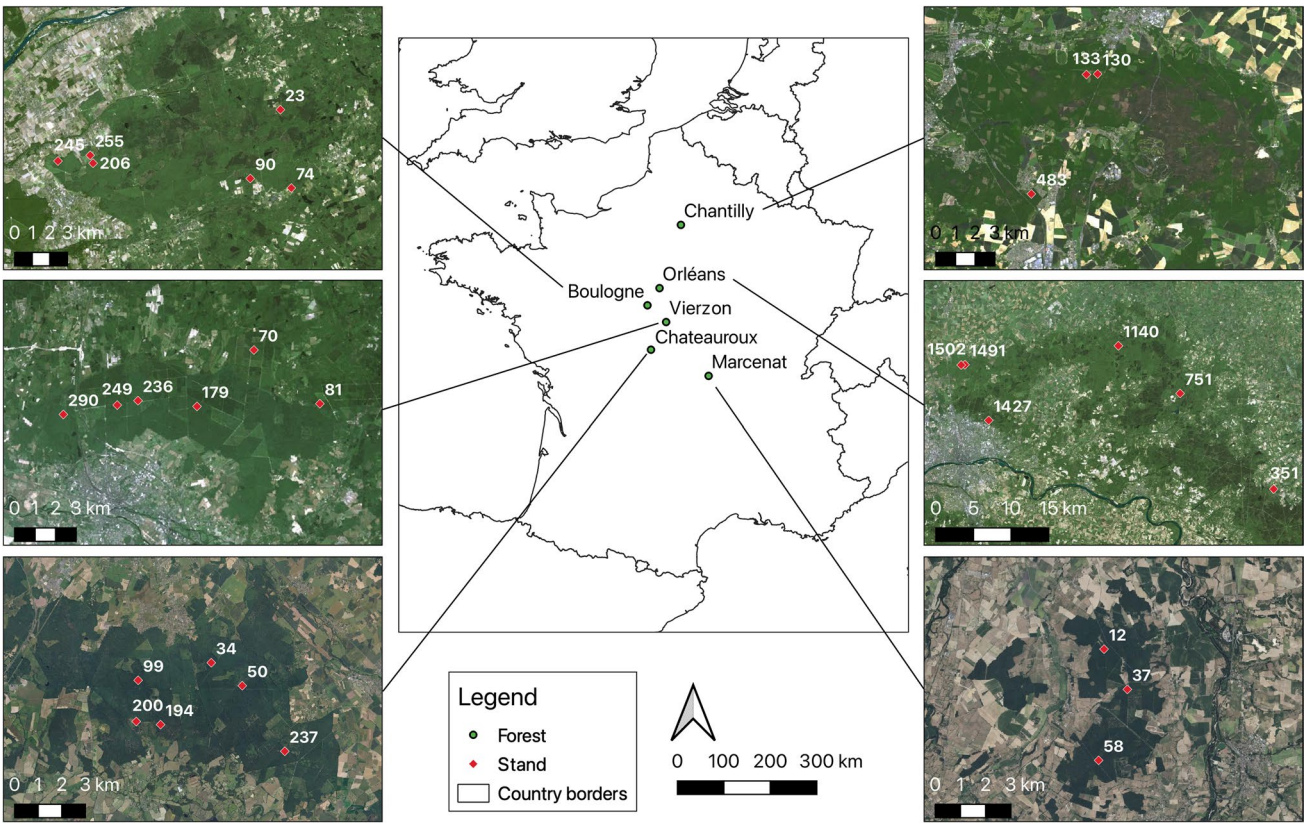


Fig. 1 Map of France showing the six forests and the selected stands, sampled between 2016 and 2021

there were slight changes in protocol between these two periods. We used green multi-funnel traps (ChemTica Internacional, San Jose, Costa Rica) with 12 fluon-coated funnels to sample the borer communities. The traps were suspended among the lower branches of the canopy (i.e. approximately 10–15 m above the ground). The collectors were filled with a 50/50% (*v/v*) solution of monopropylene glycol and water plus a drop of detergent. No lure was added to the traps. This set-up (a green trap in the canopy layer) is one of the most efficient ways to capture Agrilinae (Rassati et al. 2019;

Sallé et al. 2020), since young Agrilinae adults carry out maturation feeding and mate on foliage (Sallé et al. 2014). Each year, the traps were hung from approximately April to September and were emptied every month. From 2016 to 2018, we installed one trap per stand. From 2019 to 2021, we placed two traps per stand, one in each of two plots (see below). In 2021, a 13-mm wire mesh was added between the trap and the collector to avoid catching bats.

All collected Agrilinae were identified to the species level according to the identification keys in Farrugia (2007) and Schaefer (1949). We computed the number of species (richness) and the number of individuals (abundance) at the plot level (corresponding to one trap), either cumulated over the whole yearly sampling season or by month.

Plot and stand characteristics

We selected stands dominated by mature oaks, with a minimal surface area of 3 ha, to avoid potential edge effects. We considered environmental variables at both plot and stand scales in our study. The plot scale corresponded to the tree equipped with a trap plus the ten closest oaks. The stand scale corresponded to all of the plots within a given stand. We considered these two embedded spatial scales because both scales may affect the sampling for different reasons. The plot scale should reflect environmental conditions in the vicinity of the traps, which may affect their attractiveness. The larger stand scale should better reflect the environmental conditions under which the borer population developed. Sampling two plots per stand also allowed us to better take into account potential variability in borer populations within stands.

One plot per stand was sampled in 2016–2018. In each plot, we measured the average tree height for five trees, tree density over 2000 m^2 and total oak basal area. Two plots per stand, separated by at least 50 m, were sampled in 2019–2021. At the plot scale, the same parameters previously described were recorded, and then stand-level characteristics were calculated by averaging the parameter values of three plots: the two plots with a trap and an additional plot, with no trap.

The level of decline for each plot and stand was evaluated each year in winter, following the DEPERIS protocol (Goudet and Nageleisen 2019). This protocol quantifies decline at the tree level by evaluating the percentage of dead branches and ramification loss in the crown. The ten closest oaks surrounding each trap were surveyed from 2016 to 2018. From 2019 to 2021, 30 oaks (i.e. ten per plot) were assessed for each monitored stand. The decline level at the plot and stand levels was the proportion of declining oaks (i.e. with a DEPERIS classification ≥ D) before (i.e. winter n) and after (i.e. winter $n+1$) the beetle sampling period. Plots and stands were considered healthy when the proportion of declining oak was between 0 and 0.3, moderately declining when the proportion was between 0.3 and 0.6 and severely declining when the proportion was above 0.6.

At the plot scale, we also gathered geographical data (longitude, latitude and altitude) and climatic data. We computed mean yearly and monthly temperature and precipitation between 2015 and 2020 from the easyclimate R package (Cruz-Alonso et al. 2023). This package makes available high-resolution (1 km) daily climate data (precipitation, minimum and maximum temperatures) for Europe from the European climatic database hosted at the University of Natural Resources and Life Sciences in Vienna, Austria. Data are currently available from 1950 to 2020. Stand and plot characteristics are presented in Supplementary Table S1.

Data analysis

All analyses were performed in R, version 4.1.1 (R Core Team 2021), and graphs were made with the ggplot2 R package. The description of the community was based on the entire data set, from 2016 to 2021, and included all the sampled forests. We then separated our data into two smaller data sets to run our analyses. To investigate the factors driving the local abundance and diversity of oak borers, we used only the data collected in the six forests in 2021, i.e. when a maximum number of plots, stands and forests were investigated. To explore the temporal dynamics of the community, we used only the data collected in Vierzon between 2016 and 2021, i.e. the forest where the longest time series was available. We performed each analysis at both plot and stand scales.

We used multivariate regressions to investigate the relative effects of different groups of variables on the abundance of each species of borers and on the species richness and abundance of the community. Three groups of variables commonly associated with community ecology and forest entomology studies were considered: stand characteristics (i.e. decline level, mean tree height, oak basal area and tree density), geographical variables (i.e. latitude, longitude and altitude) and climatic variables (i.e. temperature and precipitation). We used GLMMs, fitted to the most relevant distribution to limit over-dispersion, i.e. the Poisson family (function *glmer,* lme4 R package), the negative binomial family (function *glmer.nb*, lme4 R package) or the log normal distribution (i.e. log + 1 transformed response, function *lmer,* lme4 R package). Thanks to the R package MuMIn, we built the best models through automatic model selection (function *dredge*) and model averaging (function *model.avg*, subset = delta < 2). Only significant variables ($p < 0.05$ across all the models) were retained in the best models. To limit co-linearity among variables, we sequentially removed variables with the highest variance inflation factor (VIF)

until all VIFs were below a threshold of 3 (function *VIF*, car R package). To account for repeat measurements and the spatial configuration of the sampling design, forest and stand were added as nested random effects on the intercept in the mixed models. There was collinearity between the climatic variables from the same month. Consequently, we selected the models by sequentially comparing models with each average monthly and yearly temperature with the mean yearly precipitation and then by sequentially comparing models with each average monthly and yearly precipitation with the mean yearly temperature. We kept the models that best explained our data. We used the same procedure to assess the opposite relationship, i.e. the effect of borer abundance and species richness on oak decline severity, with also stand characteristics (excluding decline), geographical and climatic variables as independent variables.

To determine whether the borer communities collected in 2021 were influenced by plot or stand decline category (i.e. healthy (< 0.3 declining trees), moderately declining (0.3 and 0.6) and severely declining (> 0.6)), we performed a PERMANOVA (function *adonis2*, Bray–Curtis index, by = "term", 999 permutations, vegan R package). We added forest as a covariate, and the forest-decline category interaction was analysed.

In order to investigate the temporal dynamics of the oak-borer community, we used the data collected in Vierzon between 2016 and 2021 with a standardised sampling period from early June to the end of August. Firstly, we tested the potential temporal autocorrelation of abundance, species richness, plot and stand decline levels with the *ACF* function. No autocorrelation was found (Fig. S1). We assessed the effect of sampling year on the abundance and species richness per trap of the overall community, on the abundance of each species separately and on the proportion of declining trees per plot or stand using a Kruskal–Wallis test. We then used a post hoc Dunn test (function *dunn_test*, Holm adjustment method, rstatix R package). To determine whether the composition of the community was driven by the above-mentioned decline category, we performed a PERMANOVA. We added sampling year as a covariate, and the interaction between sampling year and decline category was analysed. To assess whether inter-annual fluctuations were synchronous among species, we performed Spearman and Pearson correlation tests (function *cor*, stats R package) and we realised a correlogram (function *corrplot*, order = hclust, corrplot R package).

Results

Composition, distribution and phenology of the community of oak-associated borers

Composition: From 2016 to 2021, 41,919 individuals belonging to ten species of oak-associated borers were collected. Eight *Agrilus* species developing on oaks were found: *Agrilus laticornis* (Illiger) (16,188), *A. angustulus* (8662 individuals), *Agrilus hastulifer* (Ratzeburg) (5341), *Agrilus obscuricollis* Kiesenwetter (4176), *Agrilus graminis* Kiesenwetter (3472), *A. sulcicollis* (1835), *A. biguttatus* (569) and *Agrilus curtulus* Mulsant and Rey (75). We also caught two *Coraebus* species: *C. undatus* (997) and *C. fasciatus* (2), and one *Meliboeus* species: *Meliboeus fulgidicollis* (Lucas) (604). Several species associated with other host plants were also collected. Among them, the two dominant species were *Agrilus olivicolor* Kiesenwetter (20,457) and *A. viridis* var. *fagi* Ratz. (386), associated with hornbeam and beech, respectively (Jendek and Poláková 2014). We also trapped a few specimens of *Agrilus ater* (Linnaeus) (12), *Agrilus betuleti* Ratzeburg (4), *Agrilus convexicollis* Redtenbacher (9), *Agrilus cyanescens* (Ratzeburg) (1) and *Agrilus pratensis* (Ratzeburg) (1).

Distribution: Overall, the composition of the oak-borer community was similar among the six forests sampled. Nine species of oak-feeding borers were collected in all of six forests in 2021, while *A. curtulus* was found in only four (Boulogne, Chantilly, Marcenat and Vierzon) (Fig. 2). *Agrilus laticornis* dominated the community in most forests. The only exception was Marcenat, where *A. graminis* was as abundant as *A. laticornis*. Although *A. graminis* and *A. sulcicollis* were found everywhere, their relative abundance was quite variable among forests, ranging from 0.2 to 26.5% for *A. graminis* and from 0.4 to 8.9% for *A. sulcicollis*.

Phenology: Most of the borer activities were concentrated in June and July (Fig. 3), and species richness also peaked in June and July (Fig. 3). When considered separately, the abundance of all species (including *A. olivicolor* and *A. viridis*, for information) peaked either in June or in July (Fig. S2). *Agrilus viridis, A. sulcicollis, A. laticornis* and *A. angustulus* were active early in the season, while *A. biguttatus*, *A. hastulifer*, *A. graminis*, *A. curtulus*, *C. undatus* and *M. fulgidicollis* were more active during the late part of the season. *A. viridis* was the only species active during the entire sampling season, from April to October. *M. fulgidicollis, A. obscuricollis, A. hastulifer, A. curtulus* and *A. biguttatus* were active for a shorter period of time than the other oak borers.

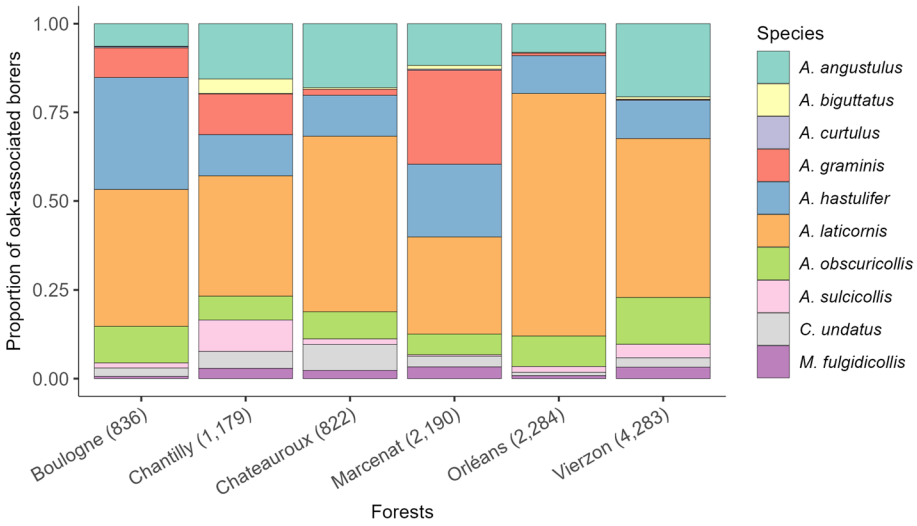


Fig. 2 Stacked diagram of the relative abundance of oak borer species, collected in 2021 in six oak forests. The total abundance of borers in each forest is given between brackets. The number of traps and the average number of insects caught per trap were as follows: Boulogne—12 traps and 70 individuals per trap; Chantilly—6 and 197; Châteauroux—12 and 69; Marcenat—5 and 438; Orléans—12 and 190; Vierzon: 12 and 357

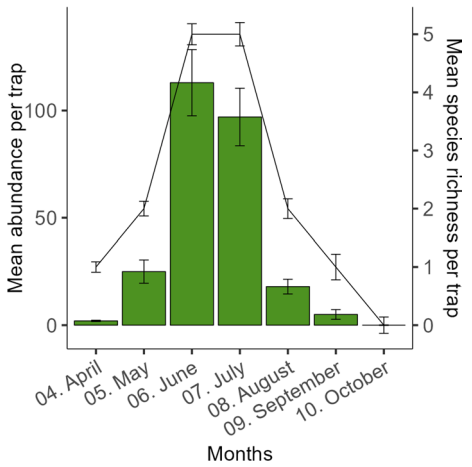


Fig. 3 Seasonal variations in abundance (bars) and richness (line) of oak borers in oak forests. Mean values (±SE) per trap, collected from 2016 to 2021 in six French oak-dominated state forests (see Table 1 for details)

Effects of local environmental variables on oak borers

Effects at the species level: Most oak borer species were promoted by oak decline, except *A. biguttatus* (Table 2). *A. olivicolor* and *A. viridis*, which do not feed on oak, were unaffected by the proportion of declining oak trees. While most monthly and annual mean temperature and precipitation values were discarded during the model selection phase, some species were favoured either by warm spring temperatures in April, May or June (*A. hastulifer, A. obscuricollis*, *A. sulcicollis*) or by high levels of precipitation in January or October (*A. biguttatus, M. fulgidicollis*). Most species were negatively impacted by stand height (except *A. angustulus, A. graminis, C. undatus* and *A. olivicolor* for which this variable was not selected). Finally, several species were promoted by longitude (*A. biguttatus, A. graminis, M. fulgidicollis*).

Effects at the community level: According to the results of average models, oak decline severity was the main factor influencing both the diversity and overall abundance of oak associated at both plot and stand scales (Table 3). Increasing tree density also promoted the abundance of borers. Most monthly mean temperature and precipitation values were discarded during the model selection phase as well as the annual mean temperature and precipitation. Only the mean temperature in June had an influence on the abundance (Table 3). Finally, longitude also influenced borer abundance at the plot level. The reverse analysis, investigating the factors influencing oak decline, indicated that the severity of decline was slightly promoted by the species richness of borers at the stand level (Table S2) but not by the overall borer abundance. At the plot level, oak decline was slightly affected by tree density (Table S2).

Since decline severity was the main driver of both the abundance and diversity of borers, we further investigated its effect as well as that of the forest sampled on the composition of the community. Community composition of the oak-dwelling borers was influenced by stand decline category ($F=2.61$, $df=2$, $R^2=0.08$, $p=0.01$) but not by forest ($F=1.81$, $df=5$, $R^2=0.12$, $p=0.06$) or the interaction between stand decline category and forest ($F=1.17$, $df=13$, $R^2=0.11$, $p=0.24$). The results at the plot level were similar.

Inter-annual dynamics of oak-associated borers in the Vierzon forest

Decline severity increased in the plots (Kruskal–Wallis, $X^2(5)=12.5$, $p=0.03$, $n=81$) and in the stands (Kruskal–Wallis, $X^2(5)=13$, $p=0.02$, $n=81$)

Table 2 Multi-model-averaged estimates for either plot- or stand-related variables determining the response of oak-associated borer species

Model	Dist. family	Scale	Predictor	Model-averaged estimate	SE	Adj. SE	*z* value	Best model	AICc
Oak-associated borers									
A. angustulus	NB	Plot	Decline	0.78**	0.30	–	2.65	Plot decline	438.90
		Stand	Decline	1.25***	0.30	–	4.16	Stand decline	438.96
A. biguttatus	P	Plot	Oak basal area	2.23***	0.50	0.51	4.35	Oak basal area + Mean January precipitation + Stand height + Tree density + Longitude	173.82
			Mean January precipitation	2.39**	0.76	0.77	3.09		
			Stand height	−1.78**	0.65	0.66	2.70		
			Tree density	−3.52***	0.92	0.94	3.75		
			Longitude	3.07***	0.66	0.67	4.56		
		Stand	–	–	–	–	–	–	239.79
A. graminis	Log	Plot	Longitude	0.80**	0.28	0.29	2.77	Longitude	181.53
			Decline	0.31*	0.14	0.14	2.20		
		Stand	Decline	0.40**	0.14	0.14	2.79	Decline + Longitude	180.29
			Longitude	0.69*	0.30	0.31	2.23		
			Tree density	0.38*	0.17	0.18	2.15		
A. hastulifer	NB	Plot	Decline	0.87**	0.29	0.29	2.99	Decline + Stand height + Mean June temperature	412.54
			Stand height	−0.80**	0.30	0.30	2.62		
			Mean June temperature	1.03***	0.24	0.25	4.14		
		Stand	Decline	1.03***	0.28	0.28	3.68	Decline + Mean annual precipitation + Mean June temperature	417.30
			Mean annual precipitation	−0.55*	0.26	0.26	2.08		
			Mean June temperature	0.83	0.25	0.26	3.21		
A. laticornis	Log	Plot	Decline	0.60**	0.19	0.20	3.05	Decline + Stand height	206.88
			Stand height	−0.53	0.25	0.25	2.08		
		Stand	Decline	0.71***	0.20	0.20	3.51	Decline + Tree density + Stand height	204.04
			Tree density	0.45*	0.20	0.20	2.30		
			Stand height	−0.42*	0.20	0.21	2.02		
A. obscuricollis	NB	Plot	Decline	0.64**	0.21	0.21	3.03	Decline + Stand height + Mean June temperature	394.39
			Stand height	−0.87***	0.25	0.25	3.42		
			Mean June temperature	0.85***	0.21	0.22	3.92		
		Stand	Decline	0.74**	0.22	0.22	3.29	Decline + Stand height + Mean June temperature	392.16
			Stand height	−0.87***	0.24	0.24	3.59		
			Mean June temperature	0.83***	0.22	0.22	3.73		
A. sulcicollis	NB	Plot	Decline	0.94***	0.27	0.28	3.38	Decline + Stand height + Mean May temperature	257.80
			Stand height	−1.29***	0.33	0.34	3.81		
			Mean May temperature	0.84**	0.28	0.28	2.95		
		Stand	Decline	0.88***	0.24	–	3.68	Decline + Stand height + Mean April temperature	256.58
			Stand height	−1.49***	0.33	–	−4.51		
			Mean April temperature	0.78**	0.28	–	2.75		
C. undatus	NB	Plot	Decline	0.94***	0.24	0.25	3.83	Decline	271.42
		Stand	Decline	0.71**	0.25	0.27	2.76	Decline	277.46
M. fulgidicollis	NB	Plot	Decline	0.77***	0.22	0.23	3.37	Decline + Mean October precipitation + Stand height + Longitude	237.85
			Mean October precipitation	0.71*	0.27	0.27	2.57		
			Stand height	−0.91**	0.29	0.30	3.09		
			Longitude	0.80***	0.22	0.22	3.60		
		Stand	Decline	0.84***	0.23	0.26	3.55	Decline + Mean October precipitation + Stand height + Longitude	236.54
			Mean October precipitation	0.74**	0.27	0.28	2.64		
			Stand height	−0.89**	0.29	0.30	2.99		
			Longitude	0.81***	0.22	0.23	3.61		

Table 2 (continued)

Model	Dist. family	Scale	Predictor	Model-averaged estimate	SE	Adj. SE	*z* value	Best model	AICc
Other borers									
A. olivicolor	Log	Plot	Mean December precipitation	−0.64	0.31	0.32	1.99	Mean December precipitation	232.14
		Stand	–	–	–	–	–	–	232.72
A. viridis	P	Plot	Oak basal area	0.89**	0.33	0.34	2.63	Oak basal area + Tree density + Latitude	202.64
			Tree density	1.01*	0.40	0.41	2.47		
			Latitude	1.68*	0.70	0.71	2.39		
		Stand	Oak basal area	−0.89*	0.39	0.40	2.22	Oak basal area + Mean December precipitation + Stand height + Tree density	215.39
			Mean December precipitation	−1.02*	0.47	0.48	2.13		
			Stand height	−1.19*	0.56	0.57	2.08		
			Tree density	−1.29*	0.63	0.65	2.00		

Generalised linear mixed-effects models (GLMMs) were fitted for the negative binomial family (NB), the Poisson family (P) or the log normal family (Log; i.e. log + 1 transformed response) with forest and stand as nested random effects. Several variables were included in the model selection process: stand characteristics (decline level, mean oak height, oak basal area and tree density), geographical variables (latitude, longitude and altitude) and climatic variables (monthly or yearly mean temperature and precipitation). Only significant variables (***$p < 0.001$, **$p < 0.01$, *$p < 0.05$) were selected. The best model was the one with the lowest AICc

SE standard error *and Adj. SE* adjusted standard error

Table 3 Multi-model-averaged estimates for either plot- or stand-related variables determining the response of oak-associated borer abundance or richness

Model	Scale	Predictor	Model-averaged estimate	SE	Adj. SE	*z* value	Best model	AICc
Abundance	Plot	Decline	0.68***	0.20	0.20	3.33	Plot decline + Longitude	697.4
		Longitude	0.42*	0.21	0.21	1.99		
		Mean June T°C	0.45*	0.22	0.23	1.99		
	Stand	Decline	0.85***	0.19	0.19	4.39	Stand decline + Tree density	689.9
		Tree density	0.53**	0.19	0.19	2.77		
Richness	Plot	Decline	0.18**	0.06	0.06	3.06	Plot decline	268.9
	Stand	Decline	0.19**	0.06	0.06	3.28	Stand decline	267.5

Generalised linear mixed-effects models (GLMMs) were fitted for the negative binomial family (abundance) or the Poisson (richness) family with forest and stand as nested random effects. Several variables were included in the model selection process: stand characteristics (decline level, mean oak height, oak basal area and tree density), geographical variables (latitude, longitude and altitude) and climatic variables (monthly or yearly mean temperature and precipitation). Only significant variables (***$p < 0.001$, **$p < 0.01$, *$p < 0.05$) were selected. The best model was the one with the lowest AICc

SE standard error, *and Adj. SE* adjusted standard error

monitored in the forest of Vierzon over the 6-year sampling period. Significant differences were observed between 2016 and 2021 at the plot scale, but no significant inter-annual variation at the stand scale was detected with the Dunn test (Fig. 4). The overall abundance of borers was influenced by sampling year (Kruskal–Wallis, $X^2(5) = 20.1$, $p = 0.001$, $n = 81$), with the largest catches occurring in 2016 and the smallest in 2017 (Fig. 4). After a sharp drop in 2017, borer abundance increased regularly from 2017 to 2021 but never regained the 2016 value. Significant differences in abundance were observed only between 2016 and 2017. Species richness was also influenced by sampling year (Kruskal–Wallis, $X^2(5) = 11.9$, $p = 0.04$, $n = 81$), although no significant inter-annual variations were detected with the Dunn test. As for the abundance, a sharp drop was observed in 2017.

At the plot scale, community composition was influenced by sampling year ($F = 2.80$, $df = 1$, $R^2 = 0.03$, $p = 0.03$) and by decline category ($F = 2.81$, $df = 2$, $R^2 = 0.07$, $p = 0.01$) but not by the interaction between sampling year and decline category ($F = 1.42$, $df = 2$, $R^2 = 0.03$, $p = 0.18$). At the stand scale, the results were slightly different. Community composition

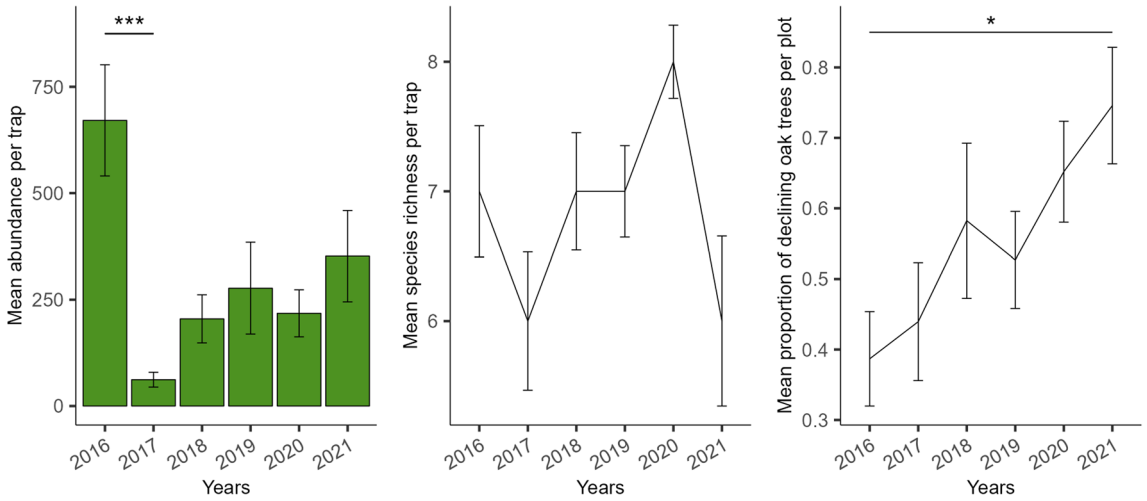


Fig. 4 Mean abundance (±SE) of oak-associated borers caught from 2016 to 2021 (left), mean species richness (±SE) of oak-associated borers caught from 2016 to 2021 (middle) and mean proportion (±SE) of declining oaks per monitored stand (right) in the Vierzon state forest. Significance levels correspond to the results of the Dunn test with ***$p<0.001$, **$p<0.01$, *$p<0.05$. Non-significant values are not shown

was influenced not only by sampling year ($F=4.09$, $df=1$, $R^2=0.05$, $p=0.006$) and decline category ($F=3.90$, $df=2$, $R^2=0.09$, $p=0.002$) but also by the interaction between sampling year and decline category ($F=2.14$, $df=2$, $R^2=0.05$, $p=0.04$).

When considered separately, the abundance of most oak-borer species experienced a marked decrease in 2017, compared to 2016 (Fig. 5). As for overall oak-borer abundance, this decrease was followed by a progressive increase during the following years for most species, and their inter-annual fluctuations were correlated, making them synchronous (Fig. S3). Only *A. graminis* showed inter-annual fluctuations that were not correlated with those of the other Agrilinae species, probably due to its peak abundance in 2020, which was not observed for the other species. Non-oak borers showed two distinct patterns. *A. olivicolor* abundance followed an inter-annual pattern similar to the oak-associated species, with a large decrease in its mean abundance per trap in 2017. In contrast, the mean abundance of *A. viridis* remained relatively constant. Only the inter-annual fluctuations of *C. undatus* (Spearman corr.=0.54, $p=0.004$), *A. laticornis* (Spearman corr.=0.34, $p=0.03$) and *M. fulgidicollis* (Spearman corr.=0.26, $p=0.04$) were positively correlated with those of oak-decline severity at the stand scale, and no correlation was observed with decline severity at the plot scale.

Discussion

Composition and phenology of the oak-borer community

During our survey, we captured all of the species of borers associated with temperate oaks in France. In terms of composition, we observed quite similar communities in the six forests sampled, where only the relative abundance of *A. graminis* and *A. sulcicollis* markedly fluctuated. The two most abundant oak-associated species collected were *A. laticornis* and *A. angustulus*. Both species are common (Farrugia 2007) and develop in oak branches (Schaefer 1949). They are consequently closely associated with the canopy, where the traps had been installed. Conversely, species developing on the trunk, such as *C. undatus*, *A. biguttatus* and *A. sulcicollis* (Hastir and Gaspar 2002; Jendek and Grebennikov 2011), were far less abundant in our traps, suggesting that canopy traps might be less efficient for these species, as suggested by Brown et al. (2017) for *A. biguttatus*. We also captured Mediterranean species, sometimes in large numbers, like *A. hastulifer* and *A. graminis*, as previously reported in Sallé et al. (2020). Until recently, these species were considered rare in the

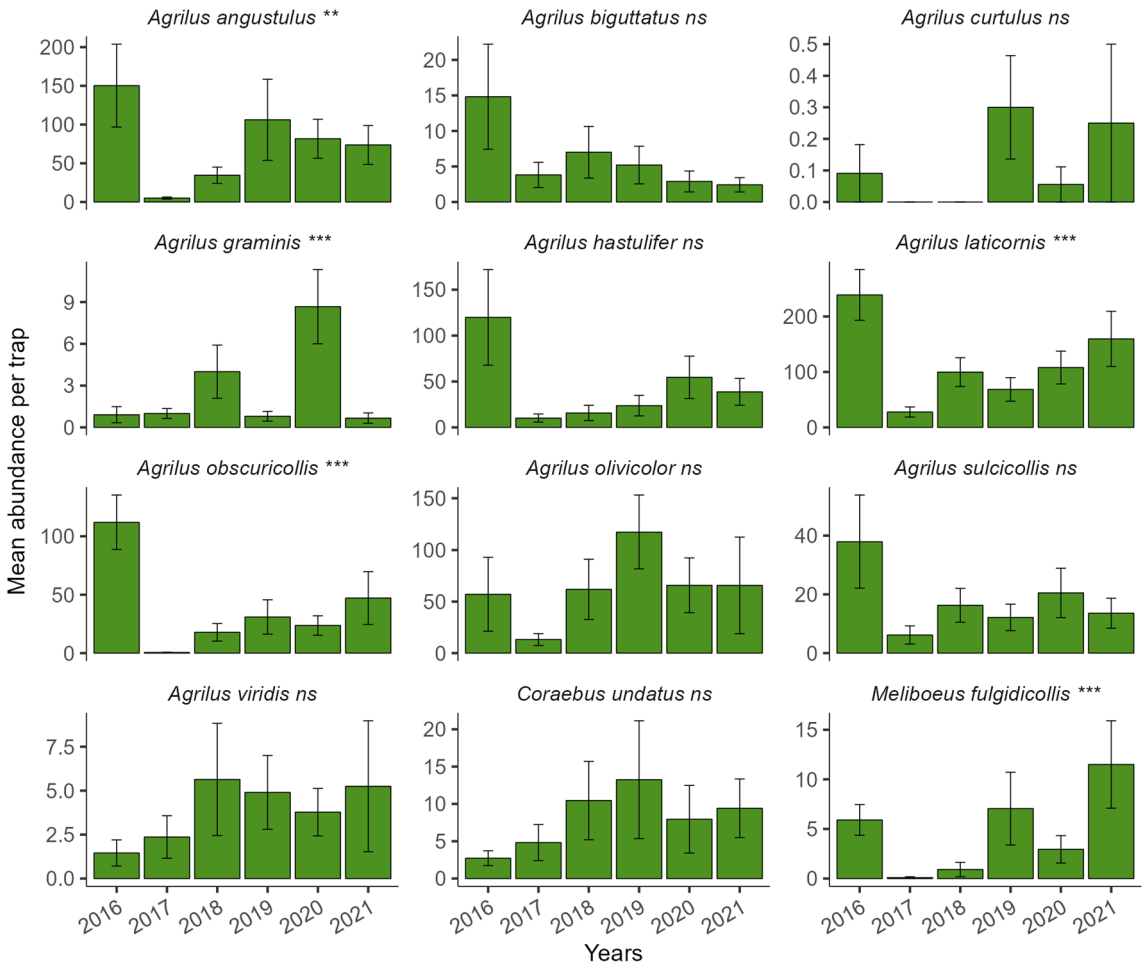


Fig. 5 Mean abundance (±SE) of oak-associated borer species caught from 2016 to 2021 in the Vierzon forest. The significance of each Kruskal–Wallis test is given after the name of the species. ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$. *Agrilus olivicolor* and *A. viridis* were added for information

area (Chapelin-Viscardi and Binon 2007), but a recent increase in abundance is in agreement with the current northwards expansion of several other Agrilinae species in Europe. Whether these abundance results relate to the number of borers that have developed locally, both in the canopy and on the trunk, or whether they better reflect flight activity at the canopy level is still unclear. Comparing results from multi-funnel traps to those from rearing traps in a same plot or stand might help to answer this question.

As we continued sampling for several years and in several forests, we were able to obtain a robust overview of the phenology of borer species in temperate oak forests. Adult activity peaked in June or July for all species, with 82% of all catches. This suggests that a sampling period ranging from mid-May until mid-August covers most of the beetles' activity period.

Forest decline drives the local abundance, richness and composition of oak-associated borers

Our results confirm that oak decline influences the local abundance, richness and community composition of oak borers. Most oak borer species were promoted by oak decline as previously reported in France (Sallé et al. 2020). Consequently, decline severity also markedly affected the composition of the oak-borer community. This is congruent with the preference of *Agrilus* spp. for weakened hosts (e.g. Dunn et al. 1987, 1990; Vansteenkiste et al. 2004; McCullough et al. 2009). Nonetheless, the absence of decline effect on the abundance of *A. biguttatus* contradicts previous observations (e.g. Sallé et al. 2014, 2020; Brown et al. 2017). The lack of decline effect might result from a less efficient sampling of this species with our

trapping design (see above). It was one of the less abundant species and was sampled in a limited number of traps.

While borers were markedly promoted by oak decline, their diversity only marginally increased stand decline severity, and their overall abundance had no influence. This is however in line with their known contribution as aggravating factors during the decline process, i.e. at the end of the process, when trees have already been severely weakened by other abiotic or biotic agents (Sallé et al. 2014; Brück-Dyckhoff et al. 2019; Macháčová et al. 2022). During our survey, oaks were sometimes severely weakened in some of our plots and stands, but there was no massive mortality. Therefore, it would be interesting to further investigate the contribution of borers to oak mortality, in unmanaged stands where declining oaks are not harvested, to evaluate whether developing management strategies would be relevant to slow down the oak decline process.

Our multi-year monitoring in the Vierzon forest showed that oak decline increased throughout the period (2016–2021), which was marked by several severe summer droughts (2018, 2019, 2020). The resulting water stress probably affected oak health. We expected that the weakening of their host trees would promote borer populations over the years. Both the species richness and abundance of oak-dwelling borers exhibited significant inter-annual fluctuations. These fluctuations were remarkably synchronous among most species, suggesting that the different species probably responded to similar environmental drivers. For several species, these inter-annual variations correlated with those of decline level. Nonetheless, the sharp decline in abundance and species richness observed in 2017 was apparently unrelated to any particular change in stand condition. Borer community composition was influenced by decline level over the monitored period and also marginally by sampling year. Although decline level influenced the population dynamics of oak borers, our results suggest that it was not the sole driver of the inter-annual variations.

Other environmental drivers involved in oak-borer population dynamics and community composition

The analysis of the inter-annual fluctuations in population levels in Vierzon indicates that the population levels of most of the borer species sharply dropped in 2017 and then slowly and progressively recovered. These fluctuations may have resulted from environmental or endogenous factors. However, since the abundance of all the species fluctuated synchronously, endogenous factors are unlikely. The drop in 2017 might have resulted from higher predation or parasitism in 2016–2017. Although we cannot totally rule this out, the magnitude of the population crash and the fact that it was observed synchronously in different species with slightly different ecological niches do not fully support this hypothesis. Another possibility is that the climatic conditions in 2017 caused the death of most of the borers. Spring 2017 in Central and Western Europe was characterised by an unusually long warm period followed by a late frost at the end of April (Vitasse and Rebetez 2018). In the Vierzon forest, temperatures between -0°C and -3°C were recorded from April 19 to 29 (data extracted from the easyclimate R package). Most Agrilinae overwinter as larvae in the cortical tissues of their host tree or in pupal cells in the outer bark or outer sapwood, and their pupation occurs in spring (e.g. Haack and Benjamin 1982; Chamorro et al. 2015). Overwintering *Agrilus* larvae can tolerate severe frosts (e.g. Crosthwaite et al. 2011). Nonetheless, the unusually warm early spring of 2017 could have broken the dormancy of the overwintering larvae, making them more vulnerable to subsequent frosts. Interestingly, other forest insects were directly affected by the late frost in 2017. In Spain, the abundance of the Asian chestnut gall wasp, *Dryocosmus kuriphilus* Yasumatsu, was reduced by 25–40%. It took three years for the population to recover from the event (Lombardero et al. 2021). Late frosts also occurred in Vierzon in 2016, 2019 and 2021, but without similar effects. However, these late frosts were not preceded by exceptionally long warm periods (Cantat et al. 2019; METEO FRANCE 2016, 2017, 2019, 2021). With climate change, springs are becoming warmer, temperatures rise earlier (Trenberth et al. 2007), and the frequency of late spring frosts is increasing in Europe (Zohner et al. 2020). Although freeze stress may affect host-tree vigour, thus promoting the colonisation by secondary pests (e.g. La Spina et al. 2013; Ranger et al. 2019) and occasionally triggering *Agrilus* outbreaks (Haack and Petrice 2019), the direct impacts of mild winters and warm springs followed by late spring frosts may hinder borer proliferation and reduce their damage in declining forests. Accordingly, while a general degradation of forest health over the last decade has been observed in France, the number of damage reports due to borers has not increased over this period (Sallé et al. 2022).

In line with these potential effects of a warm spring/late frost combination, late spring temperatures influenced the overall number of catches as well as those of several species. This is congruent with the phenology of most species (see above). Warmer spring conditions probably enhance and synchronise adult emergence and dispersal activity, leading to an increase in catches. Another climatic variable, mean precipitation in autumn or winter, also promoted the abundance of two species, *A. biguttatus* and *M. fulgidicollis*. How autumn or winter precipitation could promote the abundance of species is unclear and would require further investigation, but since both species were not very abundant, this effect should be considered with caution.

Among the variables included in our model selection, stand-related (i.e. tree density, stand height) and geographical (i.e. longitude) variables also influenced the overall

abundance of oak-dwelling borers and those of several species. Regarding stand-related variables, while increased tree density may result in cooler conditions in summer (Diaz-Calafat et al. 2023), hindering the development of thermophilous species, and decrease the attractiveness of visual traps, making them less visible when surrounded by many trees, it had a positive effect on borer abundance. In our case study, however, it seems that tree density rather acted as a reservoir for oak borers. We did not specifically measure oak density, but since our study was located in pure oak stands, an increase in tree density probably implied an increased amount of trophic resources. The detrimental impact of stand height on the abundance of most species was not expected. Since it affects trap height, stand height may reduce the accessibility or attractiveness of canopy traps. Nonetheless, stand height was rather homogeneous among stands, and this effect should consequently be considered with caution.

Finally, regarding geographical variables, our results did not support our latitude hypothesis, as latitude was discarded during the model selection process. Latitude affects Agrilinae as the fecundity, body size and weight of female *A. planipennis* are negatively correlated to latitude (Marshall et al. 2013). This is further supported by the above-mentioned northwards expansion of several other Agrilinae species in Europe promoted by climate warming (Buse et al. 2013; Brown et al. 2015), and the observed increase in abundance of Mediterranean species in our survey. We therefore expected both lower abundance and richness of oak borers at higher latitudes. On the contrary, longitude marginally influenced the abundance of several species and of the community, with a greater abundance of oak borers in the eastern forests. Since our sample did not include a wide range of longitudinal gradients, this should be considered with caution. It might reflect the fact that forest decline was more acute in the eastern forests, where recent droughts have been more severe, thereby promoting borer populations.

Conclusion

The increase in the frequency, spatial extent and severity of forest decline will likely favour the expansion and abundance of borers in Europe, which may in turn further aggravate decline. Concern for the future health of European oak forests is therefore increasing. However, our results show that the sanitary condition of the host trees is not the only driver of oak-borer population dynamics. Tree density and warm June conditions can promote the local abundance and/or activity of borers. Borers were also more abundant in the easternmost forests of our sampled area. Our results also suggest that extreme climatic events such as a warm period followed by late frosts influence these communities and could markedly affect their population dynamics. Finally, our data suggest that the use of green multi-funnel traps in the oak canopy, especially from mid-May to mid-August, makes it possible to sample a large range of oak-related species, but may underestimate the population of species developing at the trunk level.

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Authors' contributions AS and CB were involved in conceptualization and writing—review and editing; ELS, AS and CB participated in data acquisition; ELS helped in analyses and writing—original draft; and AS was involved in supervision.

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Availability of data and material Stand characteristics, geographic variables and climatic variables are publicly available in an online repository (https://doi.org/10.57745/2CXMRU).

Code availability R code is publicly available in an online repository (https://doi.org/10.57745/2CXMRU).

Declarations

Conflict of interests There are no conflicting or competing interests declared by the authors.

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Consent for publication All authors have seen and approved the manuscript and have consented to publication.

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